MODULE II :Open Stack

Designing OpenStack Cloud Architectural Consideration – OpenStack - The new data center paradigm – OpenStack logical architecture - Nova - Compute Service.

What is OpenStack ?

OpenStack is a cloud operating system that controls large pools of compute, storage, and networking resources throughout a datacenter, all managed and provisioned through APIs with common authentication mechanisms.

What is OpenStack ?

OpenStack is one of the biggest cloud computing platforms and provides an “Infrastructure-as-a-Service” solution.

The variety of different services offered allows the administrator to install and configure the environment according to his own requirements.

What is OpenStack ?

- Free and open-source cloud-computing software platform.
- Provides services for managing a Cloud environment on the fly.
- Consists of a group of interrelated projects that control pools of processing, storage, and networking resources
- Provides users methods and support to deploy virtual machines in a remote environment.
- State in OpenStack is maintained in centrally managed relational database (MySQL or MariaDB).

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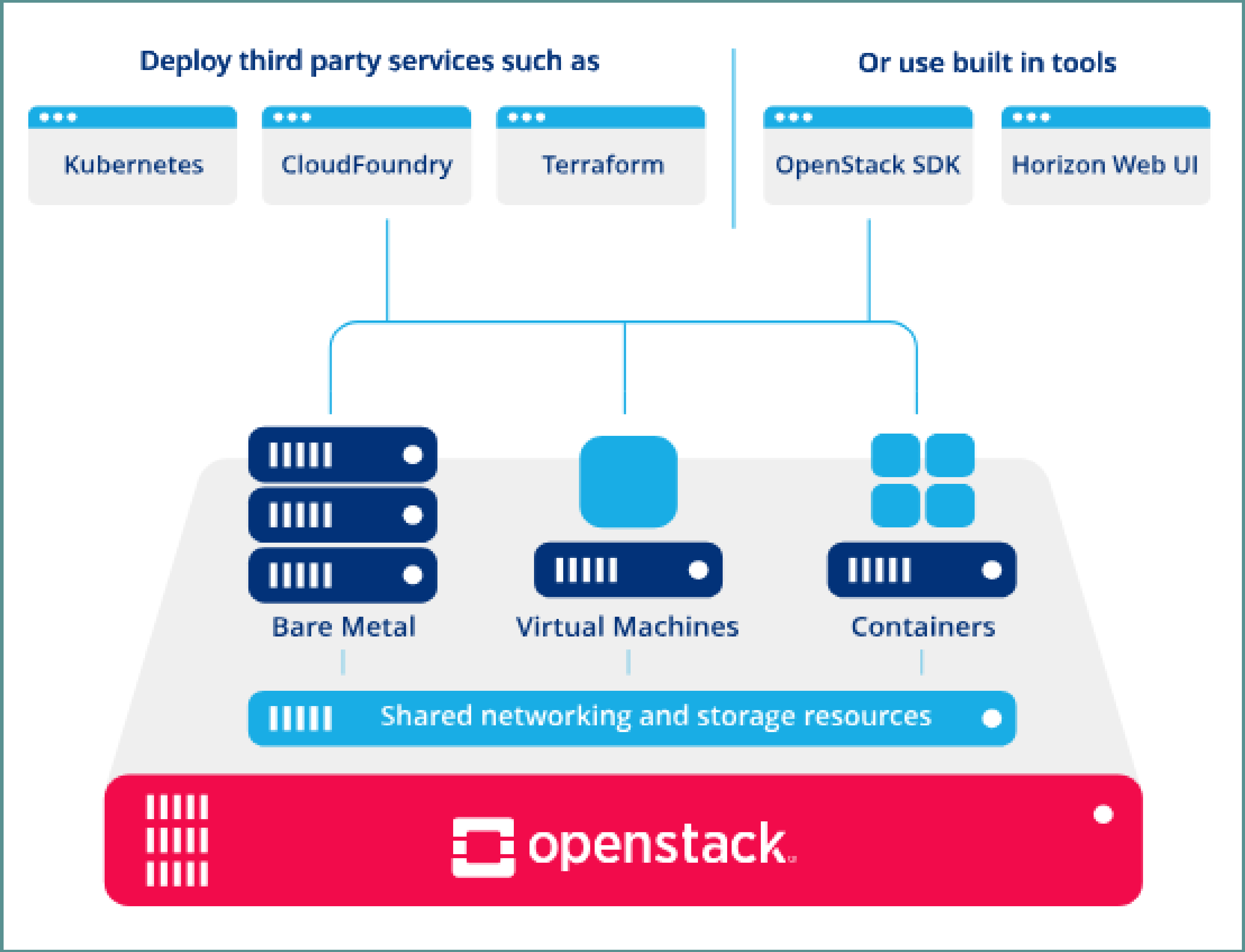
What is OpenStack ?

A dashboard is also available, giving administrators control while empowering their users to provision resources through a web interface.

What is OpenStack ?

Beyond standard infrastructure-as-a-service functionality, additional components provide orchestration, fault management and service management amongst other services to ensure high availability of user applications.


Deploy third party services such as
Or use built in tools
Kubernetes
CloudFoundry
Terraform
OpenStack SDK
Horizon Web UI
Bare Metal
Virtual Machines
Containers
Shared networking and storage resources
openstack

OPENSTACK CLOUD ARCHITECTURE

- The figure represents the high level architecture of the OpenStack cloud platform, Showing how OpenStack forms the foundation of the cloud and how users interact with it using both built-in and third party tools.
- OpenStack controls large pools of compute, storage and networking resources throughout a data center.
- These resources are managed using APIs and dashboards
- Can support different workload types such as virtual machine, containers and bare metal servers.

User Access and Management Layer(Top Layer)

- The top of the architecture represents the interfaces and tools through which users interact with OpenStack.

- Users do not directly manage servers instead, they use interfaces provided by OpenStack or external tools.

Built-in OpenStack Tools

- **Horizon Web UI**
- Horizon is the official web-based dashboard of OpenStack.
- It provides a graphical interface for users and administrators to manage cloud resources such as instances, networks, volumes and images

OpenStack SDK

- Provides APIs and libraries.
- Allows programmatic control of cloud resource.
- Provides libraries for different programming languages that allow users to interact with OpenStack services programmatically.
- Useful for automation and integration with applications.

Third Party Tools

- **Kubernetes**
- Used for container orchestration
- Runs containerized application on OpenStack infrastructure
- When integrated with OpenStack, it enables the deployment and management of containerized applications using OpenStack infrastructure

Cloud Foundry

- Is a PaaS tool that allows developers to deploy applications without managing the underlying infrastructure.
- Helps developers deploy applications easily.

Terraform

- Infrastructure as Code (IaC) tool.
- It allows users to define cloud resources in configuration files and automatically provision them on OpenStack.
- Automates the creation of VM, network and storage.
- **Purpose:** Provide flexible and multiple ways for users to request, manage and automate cloud resources

Compute Resource Layer

- This layer shows the different types of compute resources supported by OpenStack.
- **Bare Metal**
- Refers to physical servers without virtualization.
- OpenStack supports bare metal provisioning to meet high-performance and low-latency workload requirements.

Virtual Machine

- Virtualized compute instances
- Created by virtualizing physical servers and allow multiple OS to run on the same hardware.

- **Containers**

- Provide lightweight application environment.
- OpenStack supports containers either directly or through orchestration platform like kubernetes.
- OpenStack can manage bare metal, virtual machine and containers within the same cloud.

Sharing Networking and Storage Layer

- Representing the common infrastructure services shared by all compute resources.

- OpenStack networking provides:
    - -IP address management
    - -Network isolation
    - -Connectivity between instances and external network.

    - **Storage**
    - Provides multiple storage options:
    - -Block storage for virtual machines
    - -Object storage for scalable datastorage

The new data center paradigm

- The **new data center paradigm** refers to the transformation of traditional data centers into **virtualized, cloud-enabled, software-defined, and automated infrastructures**.

✓ Earlier data centers were:
- Hardware-centric
- Manually managed
- Rigid and difficult to scale

✓ Modern data centers are:
- Software-defined
- Highly automated
- Scalable and elastic
- Cloud-integrated

- This shift is driven by technologies such as virtualization, cloud computing, SDN, and containerization.

Traditional Data Center Model (Old Paradigm)

In the traditional model:
 Dedicated Hardware
Each application runs on a separate physical server.

 Manual Configuration
Network, storage, and servers are configured manually.

 Low Utilization
Servers often use only 10–20% of their capacity.

 High Capital Expenditure (CapEx)
Organizations purchase hardware in advance.

 Static Infrastructure
Scaling requires purchasing and installing new hardware.

 Silo-Based Architecture
Compute, storage, and networking teams work separately.

Limitations of Traditional Model

- Resource wastage
- Slow deployment (weeks/months)
- High maintenance cost
- Poor scalability
- Complex disaster recovery

The New Data Center Paradigm (Modern Approach)

- The new paradigm is built on **abstraction, virtualization, automation, and cloud services**.
- Instead of managing hardware directly, infrastructure is managed through software.

✓ Key Technologies Enabling the New Paradigm:

- **<u>Virtualization</u>**

Virtualization abstracts physical hardware into multiple virtual resources.
Example:
One physical server → Multiple Virtual Machines (VMs)
Hypervisors such as VMware ESXi
enable this capability.

- **Benefits:**
- Higher utilization
- Server consolidation
- Reduced hardware cost

Cloud Computing

- Cloud computing provides on-demand access to resources.

✓ **Service Models:**

- IaaS (Infrastructure as a Service)
- PaaS (Platform as a Service)
- SaaS (Software as a Service)
- Cloud platforms like OpenStack enable private cloud deployment.

✓ **Key Characteristics:**

- On-demand self-service
- Broad network access
- Resource pooling
- Rapid elasticity
- Measured service

Software-Defined Infrastructure (SDI)

- Software controls compute, storage, networking.
- API-based management.

 Software-Defined Compute

Virtual machines instead of physical servers.

 Software-Defined Storage (SDS)

Storage pooled and dynamically allocated.

 Software-Defined Networking (SDN)

Organizations like Open Networking Foundation standardize SDN principles.

- Network behavior is centrally controlled via software controllers.

✓ **Benefits:**

- Centralized control
- Network automation
- Better security policies

Automation and Orchestration

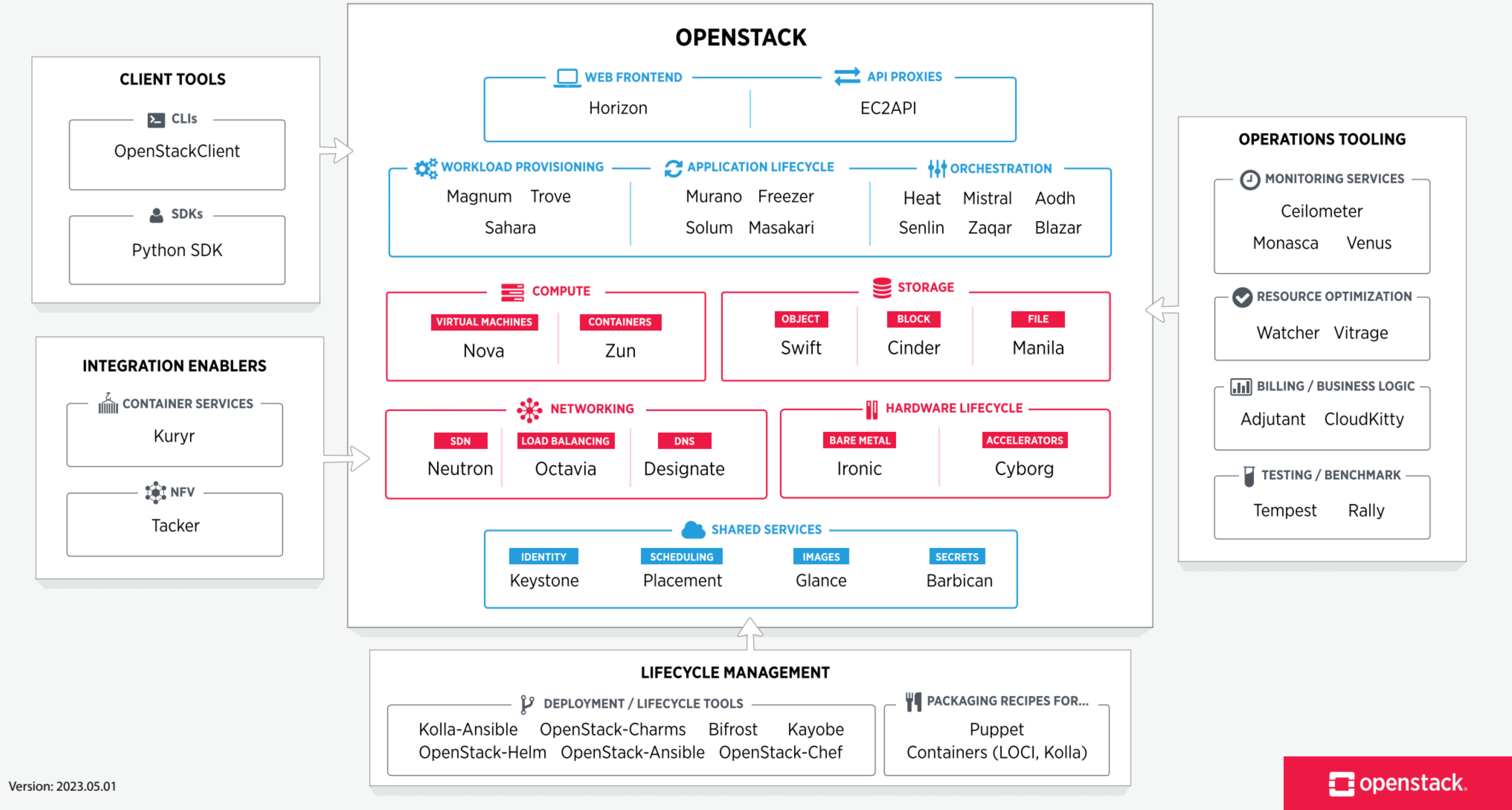
- ✓ Modern data centers use:
- • Infrastructure as Code (IaC)
- • Automated provisioning
- • Policy-based management

- ✓ This eliminates manual configuration and reduces human errors.

THE OPENSTACK LANDSCAPE

OpenStack is broken up into services to allow you to plug and play components depending on your needs.

The openstack map gives you an “at a glance” view of the openstack landscape to see where those services fit and how they can work together.



OPENSTACK
WEB FRONTEND
Horizon
API PROXIES
EC2API
WORKLOAD PROVISIONING
Magnum Trove
Sahara
APPLICATION LIFECYCLE
Murano Freezer
Solum Masakari
ORCHESTRATION
Heat Mistral Aodh
Senlin Zaqar Blazar
COMPUTE
VIRTUAL MACHINES
Nova
CONTAINERS
Zun
STORAGE
OBJECT
Swift
BLOCK
Cinder
FILE
Manila
NETWORKING
SDN
Neutron
LOAD BALANCING
Octavia
DNS
Designate
HARDWARE LIFECYCLE
BARE METAL
Ironic
ACCELERATORS
Cyborg
SHARED SERVICES
IDENTITY
Keystone
SCHEDULING
Placement
IMAGES
Glance
SECRETS
Barbican
CLIENT TOOLS
CLIs
OpenStackClient
SDKs
Python SDK
INTEGRATION ENABLERS
CONTAINER SERVICES
Kuryr
NFV
Tacker
OPERATIONS TOOLING
MONITORING SERVICES
Ceilometer
Monasca Venus
RESOURCE OPTIMIZATION
Watcher Vitrage
BILLING / BUSINESS LOGIC
Adjutant CloudKitty
TESTING / BENCHMARK
Tempest Rally
LIFECYCLE MANAGEMENT
DEPLOYMENT / LIFECYCLE TOOLS
Kolla-Ansible OpenStack-Charms Bifrost Kayobe
OpenStack-Helm OpenStack-Ansible OpenStack-Chef
PACKAGING RECIPES FOR...
Puppet
Containers (LOCI, Kolla)
Version: 2023.05.01
openstack

OpenStack components :

Core OpenStack Components include

- OpenStack services,
- Operations tooling,
- Add-Ons to Services and
- Integration enablers.

OpenStack components : Nova

- Cloud computing fabric controller, main part of an IaaS system.
- It is designed to manage and automate pools of computer resources

OpenStack components : Nova

OpenStack components : Keystone

- Provides identity services for OpenStack.
- A central list of users/permissions mapped against OpenStack services.
- Provides multiple means of access.

OpenStack components : Glance

- Provides image services to OpenStack. "images" refers to images (or virtual copies) of hard disks.
- Used as templates for deploying new VMs.

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